

Quantum Phase Estimation

1 The problem

Given a unitary U and one of its eigenstates $|\psi\rangle$, we know that

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle \quad (1)$$

for some phase $\theta \in [0, 1)$. Because U is unitary, the eigenvalue has modulus 1, so *all* the information is contained in θ . QPE estimates θ .

The computation of eigenvalues is critical for solving a vast array of problems where the QPE is the foundational primitive. Crucially, it resolves the diagonalization problem in polynomial time, as significant advantage over classical algorithms, which scale as $O(N^3)$.

QPE core components are:

- *phase kickback*, which moves the phase from the target to the control qubits;
- *inverse QFT*, which reads the phase from the control register.

2 Step 1: phase kickback

2.1 Why the phase cannot simply be measured

A global phase is unobservable. Measuring $|\psi\rangle$ after applying U gives exactly the same statistics as measuring $|\psi\rangle$ directly, because the factor $e^{2\pi i\theta}$ cancels in $|\langle x|\psi\rangle|^2$.

The trick is to turn the phase from *global* into *relative*, by putting a control qubit in superposition. This mechanism is called *phase kickback*.

2.2 How phase kickback works

Take one control qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and apply a controlled- U to $|\psi\rangle$:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|\psi\rangle \xrightarrow{cU} \frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle e^{2\pi i\theta}|\psi\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\theta}|1\rangle) \otimes |\psi\rangle \quad (2)$$

Two things are worth noticing:

1. The eigenstate $|\psi\rangle$ **factors out unchanged**: it is never disturbed, which is why the same $|\psi\rangle$ can be reused for every control qubit.
2. The phase has moved onto the *control* qubit, where it is now a relative phase and therefore measurable.

Applying U a total of k times multiplies the phase by k , since $U^k|\psi\rangle = e^{2\pi i k \theta}|\psi\rangle$, leaving the control qubit in

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i k \theta}|1\rangle) \quad (3)$$

This is the lever of the whole algorithm: using $k = 2^j$ on control qubit j writes progressively more significant bits of θ into the register.

2.3 The output state

With n control (*estimation*) qubits, each prepared in $|+\rangle$, and qubit j controlling U^{2^j} , the register ends up in

$$\frac{1}{\sqrt{2^n}} \bigotimes_{j=0}^{n-1} (|0\rangle + e^{2\pi i 2^j \theta} |1\rangle) = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i \theta k} |k\rangle \quad (4)$$

The second form deserves attention: expanding the product over all bit patterns reassembles exactly the sum, because $\sum_j 2^j k_j = k$.

That sum is precisely the **Quantum Fourier Transform (QFT) of the basis state $|2^n \theta\rangle$** :

$$\text{QFT} |y\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i y k / 2^n} |k\rangle \quad (5)$$

Matching the exponents gives $y = 2^n \theta$. The controlled- U ladder has therefore left the register holding the Fourier transform of the answer.

3 Step 2: inverse QFT to extract the phase

If the register holds $\text{QFT}|2^n \theta\rangle$, applying QFT^{-1} recovers $|2^n \theta\rangle$, and a measurement in the computational basis reads off the integer $m = 2^n \theta$, that is

$$\boxed{\theta = m/2^n} \quad (6)$$

When $2^n \theta$ is an integer m , the state before the inverse QFT is exactly $\text{QFT}|m\rangle$. The inverse QFT therefore produces $|m\rangle$ with no amplitude on any other basis state, so the measurement returns m with probability 1.

When $2^n \theta$ is *not* an integer, the inverse QFT cannot land on a single basis state and produces instead a distribution peaked at the nearest representable values:

$$P(m) = \left| \frac{\sin(\pi 2^n (\theta - m/2^n))}{2^n \sin(\pi (\theta - m/2^n))} \right|^2 \quad (7)$$

This expression is implemented as the function `expected_phase_distribution()` in the code and is overlaid on the simulated results in the following sections.

4 The circuit

The function `qpe_circuit()` assembles the three stages: the Hadamard gates, the ladder of controlled- U^{2^j} gates, and the inverse QFT. As an example, the figure below shows the circuit for the phase gate $P(\varphi)$, whose eigenstate $|1\rangle$ has eigenvalue $e^{i\varphi}$, with $\varphi = 2\pi \cdot 0.25$ and three estimation qubits.

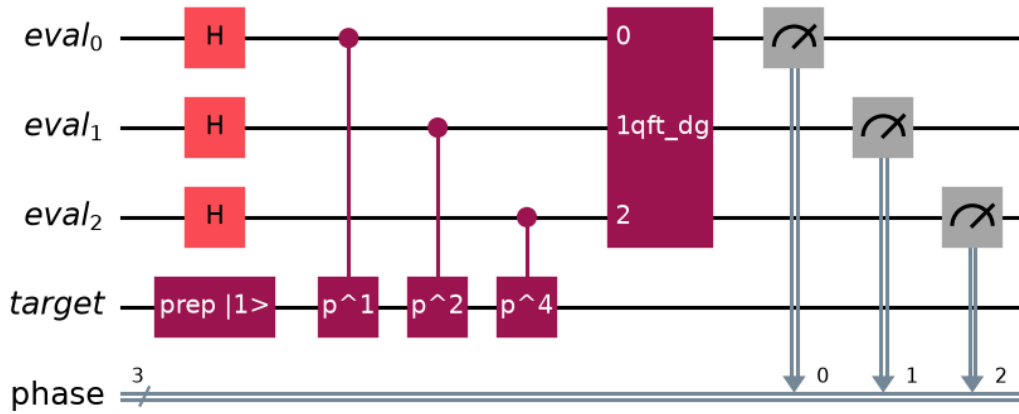


Figure 1: QPE circuit for the phase gate with 3 estimation qubits. The target qubit is prepared in the eigenstate $|1\rangle$.

The controlled powers are built by *repeating* the gate rather than by exponentiating its matrix. Both approaches are correct; repetition keeps the exponential gate count of QPE visible in the circuit, which is the honest picture of its cost, and mirrors the reference implementations.

5 Two examples: exact and non-representable phase

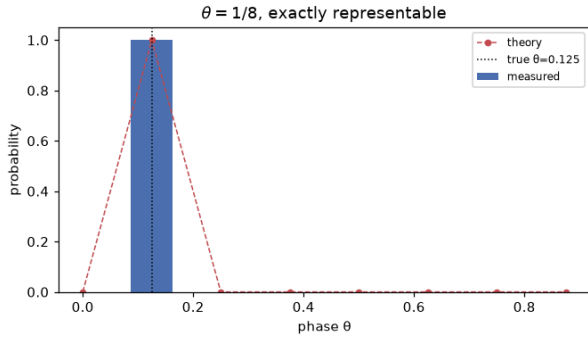
5.1 An exactly representable phase

The phase $\theta = 1/8$ needs 3 bits: $1/8 = 0.001_2$. With $n = 3$ estimation qubits, $2^n\theta = 1$ is an integer, so QPE must return $\theta = 1/8$ with certainty. Running the circuit with 8192 shots confirms it.

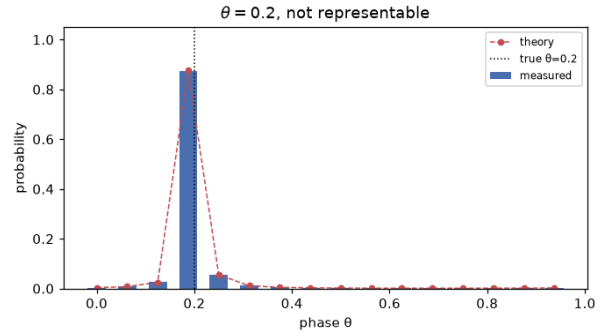
5.2 A non-representable phase

The [PennyLane tutorial](#) uses $U = R_\phi(2\pi/5)$, that is $\theta = 0.2$, with 4 estimation qubits, and observes a peak at $0.0011_2 = 0.1875$. In binary $0.2 = 0.00110011\dots$ repeats forever, so the best a 4-bit register can do is $3/16 = 0.1875$.

Quantity	Value
True phase	0.125
Estimated phase	0.125
Probability	1.0000
Error	0

Table 1: Estimation of $\theta = 1/8$ with $n = 3$.Figure 2: Measured distribution for $\theta = 1/8$.

Quantity	Value
True phase	0.2
Most likely outcome	$0.1875 = 0011_2$
Probability	0.8738
Resolution 2^{-n}	0.0625
Error	0.0125

Table 2: Estimation of $\theta = 0.2$ with $n = 4$.Figure 3: Measured distribution for $\theta = 0.2$. The red curve is the closed-form distribution.

In the exact case all the probability sits on a single outcome: $1/8$ terminates in binary within 3 bits, so the inverse QFT maps the register onto a single basis state with no leakage.

In the non-representable case the distribution is peaked but no longer a delta: the amplitude spreads over neighbouring values, with a long, rapidly decaying tail. The red curve is the closed-form expression of (7), and it lies on top of the sampled histogram, confirming that the implementation reproduces the theory quantitatively rather than just qualitatively.

The *second* most likely outcome is $0.25 = 0.0100_2$, the other neighbour of 0.2 . The distribution is asymmetric because 0.2 sits closer to 0.1875 than to 0.25 .

6 Precision: what more qubits buy

Each additional estimation qubit halves the resolution 2^{-n} , so the error falls *exponentially* in n . However, qubit j applies U a total of 2^j times, so the gate count grows exponentially as well, and the total cost is $O(2^n) = O(1/\varepsilon)$ for a target error ε .

This is the central trade-off of QPE, and the reason why “just adding more qubits” is not free. The figure below shows the estimate of $\theta = 0.2$ for n from 2 to 8, with 4096 shots each.

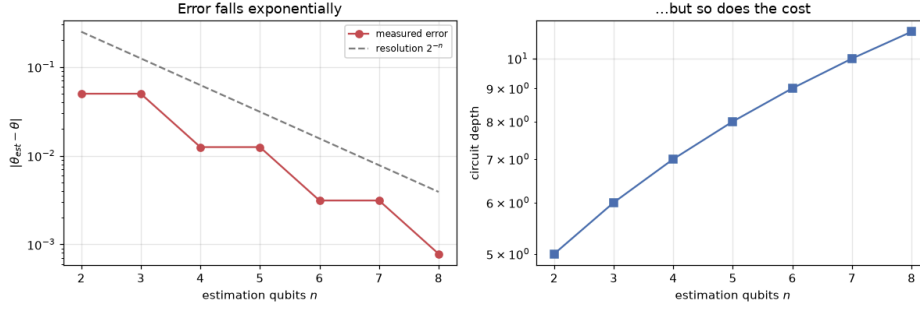


Figure 4: Left: the estimation error follows the resolution 2^{-n} . Right: the circuit depth grows as the number of estimation qubits increases.

The error tracks the resolution line closely, and the depth climbs just as fast. Buying one more bit of precision doubles the work, which is exactly why the hardware experiments keep n small.

7 Inputs that are not eigenstates

Nothing in the derivation required $|\psi\rangle$ to be an eigenstate; it only required linearity. Writing an arbitrary input in the eigenbasis of U , $|\phi\rangle = \sum_i c_i |u_i\rangle$, QPE produces

$$\sum_i c_i |\tilde{\theta}_i\rangle \otimes |u_i\rangle \quad (8)$$

so measuring the estimation register samples the eigenphase θ_i with probability $|c_i|^2$, and **collapses the target register onto the corresponding eigenstate**.

This is genuinely useful: it samples the spectrum without diagonalising anything. It is also precisely the mechanism exploited by HHL, where the input $|b\rangle$ is deliberately *not* an eigenvector of A .

As a check, consider again the phase gate with $\theta = 0.25$: $|0\rangle$ has phase 0 and $|1\rangle$ has phase θ . Preparing the target in the equal superposition $|+\rangle$ should yield both phases with equal weight, and indeed the simulation measures phase **0.000** with probability **0.5002** and phase **0.250** with probability **0.4998**.

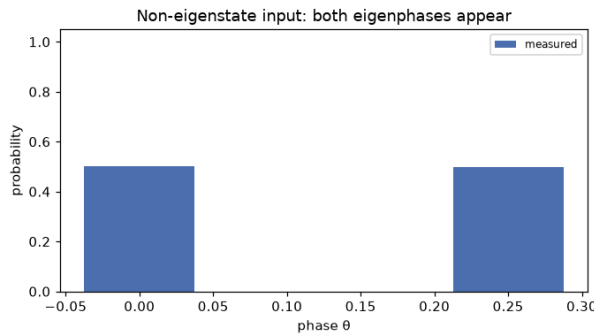


Figure 5: Measured phase distribution for the input $|+\rangle$: both eigenphases appear with equal weight.

Both eigenphases appear with weight ≈ 0.5 , matching $|c_i|^2$ for an equal superposition.